

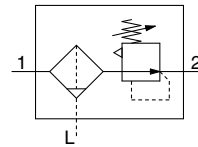
## Filter Regulator

# AW20-B to AW60-B

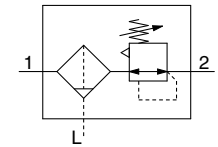
## Filter Regulator with Backflow Function

# AW20K-B to AW60K-B

Symbol  
Filter Regulator



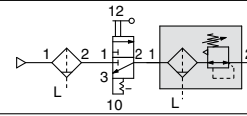
Filter Regulator with  
Backflow Function



- Integrated filter and regulator units save space and require less piping.
- With the backflow function it incorporates a mechanism to exhaust the air pressure in the outlet side reliably and quickly.

Example)

When the air supply is cut off and releasing the inlet pressure to the atmosphere, the residual pressure release of the outlet side can be ensured for a safety purpose.



## How to Order

Refer to page 85 for size 10.

AW **30** **K** - **03** **BE** - **1N** - **B**

1 2 3 4 5 6

- Option/Semi-standard: Select one each for a to i.
- Option/Semi-standard symbol: When more than one specification is required, indicate in alphanumeric order.  
Example) AW30K-03BE-1N-B

|                                            |                                | Symbol                            | Description                                                | ①                                                                         |    |                       |                       |                       |
|--------------------------------------------|--------------------------------|-----------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------|----|-----------------------|-----------------------|-----------------------|
|                                            |                                |                                   |                                                            | Body size                                                                 |    |                       |                       |                       |
|                                            |                                |                                   |                                                            | 20                                                                        | 30 | 40                    | 60                    |                       |
| ②                                          | With backflow function         | Nil                               | Without backflow function                                  |                                                                           | ●  | ●                     | ●                     | ●                     |
|                                            |                                | K <sup>Note 1)</sup>              | With backflow function                                     |                                                                           | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   |                                                            | +                                                                         |    |                       |                       |                       |
| ③                                          | Pipe thread type               | Nil                               | Rc                                                         | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                | N <sup>Note 2)</sup>              | NPT                                                        | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                | F <sup>Note 3)</sup>              | G                                                          | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                |                                   |                                                            | +                                                                         |    |                       |                       |                       |
| ④                                          | Port size                      | 01                                | 1/8                                                        | ●                                                                         | —  | —                     | —                     |                       |
|                                            |                                | 02                                | 1/4                                                        | ●                                                                         | ●  | ●                     | —                     |                       |
|                                            |                                | 03                                | 3/8                                                        | —                                                                         | ●  | ●                     | —                     |                       |
|                                            |                                | 04                                | 1/2                                                        | —                                                                         | —  | ●                     | —                     |                       |
|                                            |                                | 06                                | 3/4                                                        | —                                                                         | —  | ●                     | ●                     |                       |
|                                            |                                | 10                                | 1                                                          | —                                                                         | —  | —                     | ●                     |                       |
|                                            |                                |                                   |                                                            | +                                                                         |    |                       |                       |                       |
| ⑤                                          | a                              | Mounting                          | Nil                                                        | Without mounting option                                                   | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | B <sup>Note 5)</sup>                                       | With bracket                                                              | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | H                                                          | With set nut (for panel mount)                                            | ●  | ●                     | ●                     | —                     |
|                                            |                                |                                   |                                                            |                                                                           | +  |                       |                       |                       |
|                                            | b                              | Float type auto drain             | Nil                                                        | Without auto drain                                                        | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | C <sup>Note 6)</sup>                                       | N.C. (Normally closed) Drain port is closed when pressure is not applied. | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | D <sup>Note 7)</sup>                                       | N.O. (Normally open) Drain port is open when pressure is not applied.     | —  | ●                     | ●                     | ●                     |
|                                            |                                |                                   |                                                            |                                                                           | +  |                       |                       |                       |
|                                            | c                              | Pressure gauge <sup>Note 8)</sup> | Nil                                                        | Without pressure gauge                                                    | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | E                                                          | Square embedded type pressure gauge (with limit indicator)                | ●  | ●                     | ●                     | ●                     |
| G                                          |                                |                                   | Round type pressure gauge (with limit indicator)           | ●                                                                         | ●  | ●                     | ●                     |                       |
| M                                          |                                |                                   | Round type pressure gauge (with color zone)                | ●                                                                         | ●  | ●                     | ●                     |                       |
| Digital pressure switch <sup>Note 9)</sup> |                                | E1                                | Output: NPN output / Electrical entry: Wiring bottom entry | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                | E2                                | Output: NPN output / Electrical entry: Wiring top entry    | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                | E3                                | Output: PNP output / Electrical entry: Wiring bottom entry | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                |                                   |                                                            | +                                                                         |    |                       |                       |                       |
| ⑥                                          | d                              | Set pressure <sup>Note 10)</sup>  | Nil                                                        | 7.3 to 123 psi (0.05 to 0.85 MPa) setting                                 | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | 1                                                          | 2.9 to 29 psi (0.02 to 0.2 MPa) setting                                   | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   |                                                            |                                                                           | +  |                       |                       |                       |
|                                            | e                              | Bowl <sup>Note 11)</sup>          | Nil                                                        | Polycarbonate bowl                                                        | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | 2                                                          | Metal bowl                                                                | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | 6                                                          | Nylon bowl                                                                | ●  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | 8                                                          | Metal bowl with level gauge                                               | —  | ●                     | ●                     | ●                     |
|                                            |                                |                                   | C                                                          | With bowl guard                                                           | ●  | — <sup>Note 12)</sup> | — <sup>Note 12)</sup> | — <sup>Note 12)</sup> |
|                                            |                                |                                   | 6C                                                         | Nylon bowl with bowl guard                                                | ●  | — <sup>Note 13)</sup> | — <sup>Note 13)</sup> | — <sup>Note 13)</sup> |
|                                            |                                |                                   |                                                            |                                                                           | +  |                       |                       |                       |
| f                                          | Drain port <sup>Note 14)</sup> | Nil                               | With drain cock                                            | ●                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                | J <sup>Note 15)</sup>             | Drain guide 1/8                                            | ●                                                                         | —  | —                     | —                     |                       |
|                                            |                                |                                   | Drain guide 1/4                                            | —                                                                         | ●  | ●                     | ●                     |                       |
|                                            |                                | W <sup>Note 16)</sup>             | Drain cock with barb fitting                               | —                                                                         | ●  | ●                     | ●                     |                       |

# Filter Regulator Series AW20-B to AW60-B

## Filter Regulator with Backflow Function Series AW20K-B to AW60K-B



AW20-B, AW20K-B AW40-B, AW40K-B

|   | Symbol | Description       | ①                      |                                                                                   |                       |                       |
|---|--------|-------------------|------------------------|-----------------------------------------------------------------------------------|-----------------------|-----------------------|
|   |        |                   | Body size              |                                                                                   |                       |                       |
|   |        |                   | 20                     | 30                                                                                | 40                    | 60                    |
| ⑥ | g      | Exhaust mechanism | Nil                    | Relieving type                                                                    | ●                     | ●                     |
|   |        |                   | N                      | Non-relieving type                                                                | ●                     | ●                     |
|   | +      |                   |                        |                                                                                   |                       |                       |
|   | h      | Flow direction    | Nil                    | Flow direction: Left to right                                                     | ●                     | ●                     |
|   |        |                   | R                      | Flow direction: Right to left                                                     | ●                     | ●                     |
|   | +      |                   |                        |                                                                                   |                       |                       |
|   | i      | Pressure unit     | Nil                    | Name plate, caution plate for bowl, and pressure gauge in imperial units: MPa     | ●                     | ●                     |
|   |        |                   | Z <sup>Note 17)</sup>  | Name plate, caution plate for bowl, and pressure gauge in imperial units: psi, °F | ○ <sup>Note 19)</sup> | ○ <sup>Note 19)</sup> |
|   |        |                   | ZA <sup>Note 18)</sup> | Digital pressure switch: With unit conversion function                            | △ <sup>Note 20)</sup> | △ <sup>Note 20)</sup> |

Note 1) Set the inlet pressure to at least 7.3 psi (0.05 MPa) higher than the set pressure.

Note 2) Drain guide is NPT1/8 (applicable to the AW20(K)-B) and NPT1/4 (applicable to the AW30(K)-B to AW60(K)-B). The auto drain port comes with ø3/8" One-touch fitting (applicable to the AW30(K)-B to AW60(K)-B).

Note 3) Drain guide is G1/8 (applicable to the AW20(K)-B) and G1/4 (applicable to the AW30(K)-B to AW60(K)-B).

Note 4) Option B, G, H, M are not assembled and supplied loose at the time of shipment.

Note 5) Assembly of a bracket and set nuts (applicable to the AW20(K)-B to AW40(K)-B). Including 2 mounting screws for the AW60(K)-B

Note 6) When pressure is not applied, condensate which does not start the auto drain mechanism will be left in the bowl. Releasing the residual condensate before ending operations for the day is recommended.

Note 7) If the compressor is small (0.75 kW, discharge flow is less than 3.5 scfm (100 L/min[ANR])), air leakage from the drain cock may occur during start of operations. N.C. type is recommended.

Note 8) When the pressure gauge is attached, a 145 psi (1.0 MPa) pressure gauge will be fitted for standard (123 psi [0.85 MPa]) type. 58 psi (0.4 MPa) pressure gauge for 2.9 psi (0.2 MPa) type.

Note 9) When choosing with H (panel mount), the installation space for lead wires will not be secured. In this case, select "wiring top entry" for the electrical entry. (Select "wiring bottom entry" when the semi-standard Y is chosen simultaneously.)

Note 10) Pressure can be set higher than the specification pressure in some cases, but use pressure within the specification range.

Note 11) Refer to Chemical data on page 90 for chemical resistance of the bowl.

Note 12) A bowl guard is provided as standard equipment (polycarbonate).

Note 13) A bowl guard is provided as standard equipment (nylon).

Note 14) The combination of float type auto drain: C and D is not available.

Note 15) Without a valve function

Note 16) The combination of metal bowl: 2 and 8 is not available.

Note 17) For pipe thread type: NPT. This product is for overseas use only according to the new Measurement Law. (The SI unit type is provided for use in Japan.) Cannot be used with M: Round pressure gauge (with color zone). Available by request for special. The digital pressure switch will be equipped with the unit conversion function, setting to psi initially.

Note 18) For options: E1, E2, E3, E4. This product is for overseas use only according to the new Measurement Law. (The SI unit is provided for use in Japan.)

Note 19) ○: For pipe thread type: NPT only

Note 20) △: Select with options: E1, E2, E3, E4.

### Standard Specifications

| Model                                            | AW20-B                                      | AW30-B                   | AW40-B        | AW40-06-B | AW60-B |
|--------------------------------------------------|---------------------------------------------|--------------------------|---------------|-----------|--------|
| Port size                                        | 1/8, 1/4                                    | 1/4, 3/8                 | 1/4, 3/8, 1/2 | 3/4       | 3/4, 1 |
| Pressure gauge port size <sup>Note 1)</sup>      | 1/8                                         |                          |               |           |        |
| Fluid                                            | Air                                         |                          |               |           |        |
| Ambient and fluid temperature <sup>Note 2)</sup> | 23 to 140°F (−5 to 60°C) (with no freezing) |                          |               |           |        |
| Proof pressure                                   | 218 psi (1.5 MPa)                           |                          |               |           |        |
| Maximum operating pressure                       | 145 psi (1.0 MPa)                           |                          |               |           |        |
| Set pressure range                               | 7.3 to 123 psi (0.05 to 0.85 MPa)           |                          |               |           |        |
| Nominal filtration rating                        | 5 μm                                        |                          |               |           |        |
| Drain capacity (cm³)                             | 8                                           | 25                       | 45            |           |        |
| Bowl material                                    | Polycarbonate                               |                          |               |           |        |
| Bowl guard                                       | Semi-standard (Steel)                       | Standard (Polycarbonate) |               |           |        |
| Construction                                     | Relieving type                              |                          |               |           |        |
| Weight (kg)                                      | 0.20                                        | 0.36                     | 0.66          | 0.72      | 2.05   |

Note 1) Pressure gauge connection threads are not available for F.R.L. unit with a square embedded type pressure gauge or with a digital pressure switch.

Note 2) 23 to 122°F (−5 to 50°C) for the products with the digital pressure switch

# Series AW20-B to AW60-B

## Series AW20K-B to AW60K-B

### Options/Part No.

| Optional specifications                    |                                                    |                                         | Model                                             |             |                                                  |              |                      |  |
|--------------------------------------------|----------------------------------------------------|-----------------------------------------|---------------------------------------------------|-------------|--------------------------------------------------|--------------|----------------------|--|
|                                            |                                                    |                                         | AW20(K)-B                                         | AW30(K)-B   | AW40(K)-B                                        | AW40(K)-06-B | AW60(K)-B            |  |
| Bracket assembly <sup>Note 1)</sup>        |                                                    |                                         | AW23P-270AS                                       | AR33P-270AS | AR43P-270AS                                      |              | AW62P-270AS          |  |
| Set nut                                    |                                                    |                                         | AR23P-260S                                        | AR33P-260S  | AR43P-260S                                       |              | — <sup>Note 2)</sup> |  |
| Pressure gauge                             | Round type <sup>Note 3)</sup>                      | Standard                                | G36-10-□01                                        |             | G46-10-□01                                       |              |                      |  |
|                                            |                                                    | 2.9 to 29 psi (0.02 to 0.2 MPa) setting | G36-4-□01                                         |             | G46-4-□01                                        |              |                      |  |
|                                            | Round type <sup>Note 3)</sup><br>(with color zone) | Standard                                | G36-10-□01-L                                      |             | G46-10-□01-L                                     |              |                      |  |
|                                            |                                                    | 2.9 to 29 psi (0.02 to 0.2 MPa) setting | G36-4-□01-L                                       |             | G46-4-□01-L                                      |              |                      |  |
|                                            | Square embedded type <sup>Note 4)</sup>            | Standard                                | GC3-10AS [GC3P-010AS (Pressure gauge cover only)] |             |                                                  |              |                      |  |
|                                            |                                                    | 2.9 to 29 psi (0.02 to 0.2 MPa) setting | GC3-4AS [GC3P-010AS (Pressure gauge cover only)]  |             |                                                  |              |                      |  |
| Digital pressure switch <sup>Note 5)</sup> |                                                    |                                         | NPN output: Wiring bottom entry                   |             | ISE35-N-25-MLA [ISE35-N-25-M (Switch body only)] |              |                      |  |
|                                            |                                                    |                                         | NPN output: Wiring top entry                      |             | ISE35-R-25-MLA [ISE35-R-25-M (Switch body only)] |              |                      |  |
|                                            |                                                    |                                         | PNP output: Wiring bottom entry                   |             | ISE35-N-65-MLA [ISE35-N-65-M (Switch body only)] |              |                      |  |
|                                            |                                                    |                                         | PNP output: Wiring top entry                      |             | ISE35-R-65-MLA [ISE35-R-65-M (Switch body only)] |              |                      |  |

Note 1) Assembly of a bracket and set nuts. Including 2 mounting screws for the AW60(K)-B

Note 2) Please consult with SMC regarding the set nuts for the AW60(K)-B.

Note 3) □ in part numbers for a round pressure gauge indicates a pipe thread type.

No indication is necessary for R; however, indicate N for NPT.

Please contact SMC regarding the pressure gauge supply for psi unit specifications.

Note 4) Including one O-ring and 2 mounting screws.

[ ]: Pressure gauge cover only

Note 5) In addition to the pressure switch body, lead wire with connector (2 m), adapter, lock pin, O-ring (1 pc.), mounting screw (2 pcs.) are attached.

[ ]: Switch body only. (Regarding how to order the digital pressure switch, refer to **the WEB catalog** or the Best Pneumatics No.6.)

A pressure switch can be mounted on the AW60(K)-B, with a special mounting adapter (Pressure switch adapter assembly: AW63P-310AS) and mounting screws (M3 x 0.5 x 14) which are delivered with the mounting adapter.

### Bowl Assembly/Part No.

| Bowl material      | Drain discharge mechanism                         | Drain port                                | Other            | Model      |            |            |           |        |
|--------------------|---------------------------------------------------|-------------------------------------------|------------------|------------|------------|------------|-----------|--------|
|                    |                                                   |                                           |                  | AW20-B     | AW30-B     | AW40-B     | AW40-06-B | AW60-B |
| Polycarbonate bowl | Manual discharge                                  | With drain cock                           | —                | C2SF-A     | —          | —          |           |        |
|                    |                                                   |                                           | With bowl guard  | C2SF-C-A   | C3SF-A     | C4SF-A     |           |        |
|                    |                                                   | Drain cock with barb fitting              | With bowl guard  | —          | C3SF-W-A   | C4SF-W-A   |           |        |
|                    |                                                   | With drain guide (without valve function) | —                | C2SF□-J-A  | —          | —          |           |        |
|                    | With bowl guard                                   |                                           | C2SF□-CJ-A       | C3SF□-J-A  | C4SF□-J-A  |            |           |        |
|                    | Automatic discharge <sup>Note)</sup> (Auto drain) | Normally closed (N.C.)                    | —                | AD27-A     | —          | —          |           |        |
|                    |                                                   |                                           | With bowl guard  | AD27-C-A   | AD37□-A    | AD47□-A    |           |        |
|                    |                                                   | Normally open (N.O.)                      | With bowl guard  | —          | AD38□-A    | AD48□-A    |           |        |
| Nylon bowl         | Manual discharge                                  | With drain cock                           | —                | C2SF-6-A   | —          | —          |           |        |
|                    |                                                   |                                           | With bowl guard  | C2SF-6C-A  | C3SF-6-A   | C4SF-6-A   |           |        |
|                    |                                                   | Drain cock with barb fitting              | With bowl guard  | —          | C3SF-6W-A  | C4SF-6W-A  |           |        |
|                    |                                                   | With drain guide (without valve function) | —                | C2SF□-6J-A | —          | —          |           |        |
|                    | With bowl guard                                   |                                           | C2SF□-6CJ-A      | C3SF□-6J-A | C4SF□-6J-A |            |           |        |
|                    | Automatic discharge <sup>Note)</sup> (Auto drain) | Normally closed (N.C.)                    | —                | AD27-6-A   | —          | —          |           |        |
|                    |                                                   |                                           | With bowl guard  | AD27-6C-A  | AD37□-6-A  | AD47□-6-A  |           |        |
|                    |                                                   | Normally open (N.O.)                      | With bowl guard  | —          | AD38□-6-A  | AD48□-6-A  |           |        |
| Metal bowl         | Manual discharge                                  | With drain cock                           | —                | C2SF-2-A   | C3SF-2-A   | C4SF-2-A   |           |        |
|                    |                                                   |                                           | With level gauge | —          | C3LF-8-A   | C4LF-8-A   |           |        |
|                    |                                                   | With drain guide (without valve function) | —                | C2SF□-2J-A | C3SF□-2J-A | C4SF□-2J-A |           |        |
|                    |                                                   |                                           | With level gauge | —          | C3LF□-8J-A | C4LF□-8J-A |           |        |
|                    | Automatic discharge <sup>Note)</sup> (Auto drain) | Normally closed (N.C.)                    | —                | AD27-2-A   | AD37□-2-A  | AD47□-2-A  |           |        |
|                    |                                                   |                                           | With level gauge | —          | AD37□-8-A  | AD47□-8-A  |           |        |
|                    |                                                   | Normally open (N.O.)                      | —                | —          | AD38□-2-A  | AD48□-2-A  |           |        |
|                    |                                                   |                                           | With level gauge | —          | AD38□-8-A  | AD48□-8-A  |           |        |

Note) Minimum operating pressure: N.O. type—15 psi (0.1 MPa) (AD38-A, AD48-A); N.C. type—15 psi (0.1 MPa) (AD27-A) and 22 psi (0.15 MPa) (AD37-A, AD47-A).

Bowl assembly comes with a bowl seal.

□ in bowl assembly part numbers indicates a pipe thread type (applicable tubing for auto drain).

No indication is necessary for Rc thread; however, indicate N for NPT thread, and F for G thread. (For auto drain, Nil: ø10, N: ø3/8")

Please consult with SMC separately for psi and °F unit display specifications.

## ⚠ Specific Product Precautions

Be sure to read this before handling. Refer to the back cover for Safety Instructions, "Handling Precautions for SMC Products" and the Operation Manual for F.R.L. Precautions, <http://www.smcworld.com>

### Design/Selection

#### ⚠ Warning

- Residual pressure disposal (outlet pressure removal) is not possible for the AW20-B to AW60-B even though the inlet pressure is exhausted. When the residual pressure disposal is performed, use the filter regulator with backflow function (AW20K-B to AW60K-B).
- The standard bowl for the air filter, filter regulator, and lubricator, as well as the sight dome for the lubricator are made of polycarbonate. Do not use in an environment where they are exposed to or come in contact with organic solvents, chemicals, cutting oil, synthetic oil, alkali, and thread lock solutions.

Effects of atmosphere of organic solvents and chemicals, and where these elements are likely to adhere to the equipment.

Chemical data for substances causing degradation (Reference)

| Type              | Chemical name                                                                                                     | Application examples                                                  | Material      |       |
|-------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------|-------|
|                   |                                                                                                                   |                                                                       | Polycarbonate | Nylon |
| Acid              | Hydrochloric acid<br>Sulfuric acid, Phosphoric acid<br>Chromic acid                                               | Acid washing liquid for metals                                        | △             | ×     |
| Alkaline          | Sodium hydroxide (Caustic soda)<br>Potash<br>Calcium hydroxide (Slack lime)<br>Ammonia water<br>Carbonate of soda | Degreasing of metals<br>Industrial salts<br>Water-soluble cutting oil | ×             | ○     |
| Inorganic salts   | Sodium sulfide<br>Sulfate of potash<br>Sulfate of soda                                                            | —                                                                     | ×             | △     |
| Chlorine solvents | Carbon tetrachloride<br>Chloroform<br>Ethylene chloride<br>Methylene chloride                                     | Cleansing liquid for metals<br>Printing ink<br>Dilution               | ×             | △     |
| Aromatic series   | Benzene<br>Toluene<br>Paint thinner                                                                               | Coatings<br>Dry cleaning                                              | ×             | △     |
| Ketone            | Acetone<br>Methyl ethyl ketone<br>Cyclohexane                                                                     | Photographic film<br>Dry cleaning<br>Textile industries               | ×             | ×     |
| Alcohol           | Ethyl alcohol<br>IPA<br>Methyl alcohol                                                                            | Antifreeze<br>Adhesives                                               | △             | ×     |
| Oil               | Gasoline<br>Kerosene                                                                                              | —                                                                     | ×             | ○     |
| Ester             | Phthalic acid dimethyl<br>Phthalic acid diethyl<br>Acetic acid                                                    | Synthetic oil<br>Anti-rust additives                                  | ×             | ○     |
| Ether             | Methyl ether<br>Ethyl ether                                                                                       | Brake oil additives                                                   | ×             | ○     |
| Amino             | Methyl amino                                                                                                      | Cutting oil<br>Brake oil additives<br>Rubber accelerator              | ×             | ×     |
| Others            | Thread-lock fluid<br>Seawater<br>Leak tester                                                                      | —                                                                     | ×             | △     |

○: Essentially safe △: Some effects may occur. ×: Effects will occur.

When the above factors are present, or there is some doubt, use a metal bowl for safety.

### Maintenance

#### ⚠ Warning

- Replace the element every 2 years or when the pressure drop becomes 0.1 MPa, whichever comes first, to prevent damage to the element.

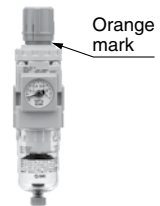
### Mounting/Adjustment

#### ⚠ Warning

- Set the regulator while verifying the displayed values of the inlet and outlet pressure gauges. Turning the regulator knob excessively can cause damage to the internal parts.
- Do not use tools on the pressure regulator knob as this may cause damage. It must be operated manually.

#### ⚠ Caution

- Be sure to unlock the knob before adjusting the pressure and lock it after setting the pressure. Failure to follow this procedure can cause damage to the knob and the outlet pressure may fluctuate.
  - Pull the pressure regulator knob to unlock. (You can visually verify this with the "orange mark" that appears in the gap.)
  - Push the pressure regulator knob to lock. When the knob is not easily locked, turn it left and right a little and then push it (when the knob is locked, the "orange mark", i.e., the gap will disappear).
- A knob cover is available to prevent careless operation of the knob. Refer to page 97 for details.
- When the bowl is installed on the AW30-B to AW60-B, install them so that the lock button lines up to the groove of the front (or the back) of the body to avoid drop or damage of the bowl.



Orange mark



Lock button

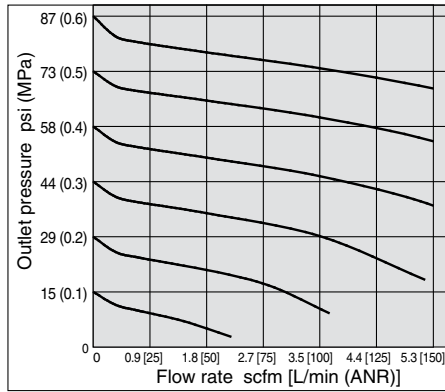
# Series AW210-A / Series AW20-B to AW60-B / Series AW20K-B to AW60K-B

## Flow-rate Characteristics (Representative values)

Condition: Inlet pressure 102 psi (0.7 MPa)

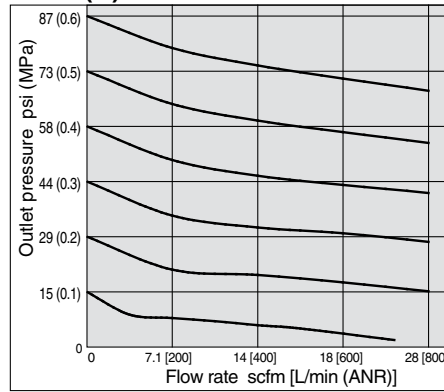
**AW10-A**

M5



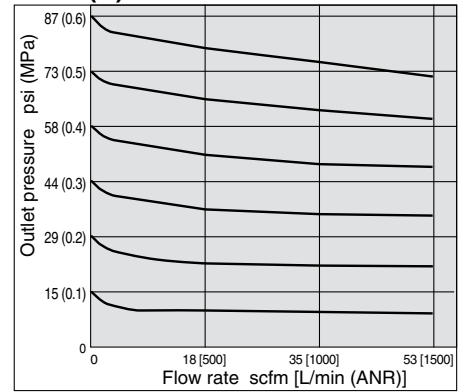
**AW20(K)-B**

Rc1/4



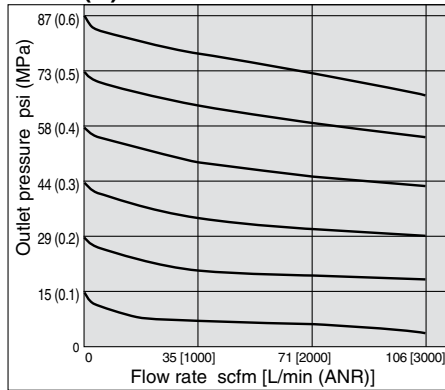
**AW30(K)-B**

Rc3/8



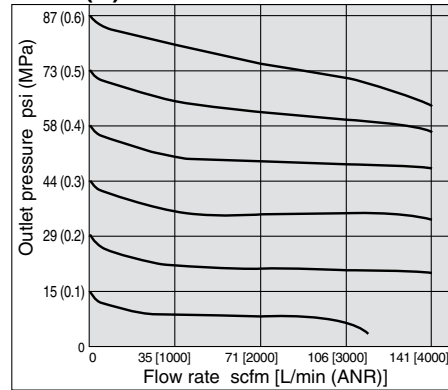
**AW40(K)-B**

Rc1/2



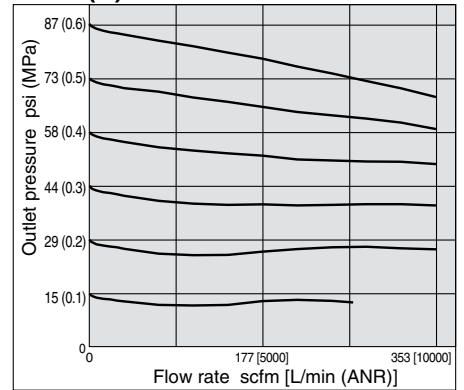
**AW40(K)-06-B**

Rc3/4



**AW60(K)-B**

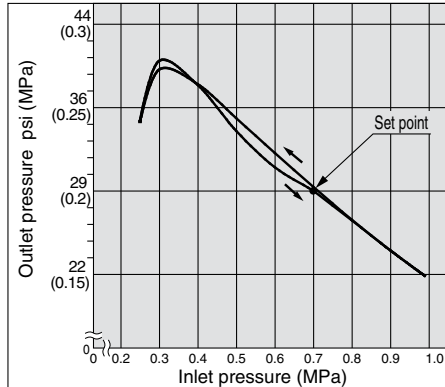
Rc1



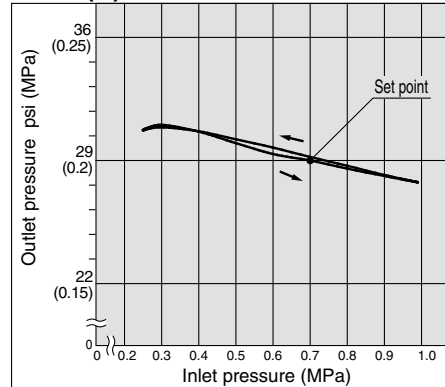
## Pressure Characteristics (Representative values)

Conditions: Inlet pressure 102 psi (0.7 MPa), Outlet pressure 29 psi (0.2 MPa), Flow rate 0.7 scfm (20 L/min) (ANR)

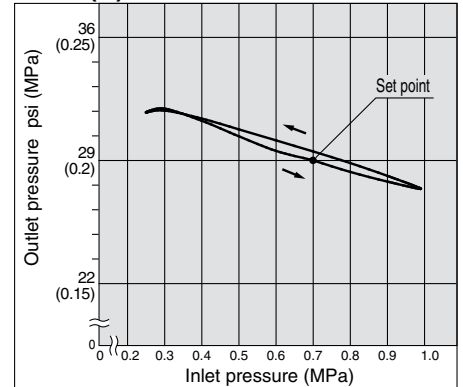
**AW10-A**



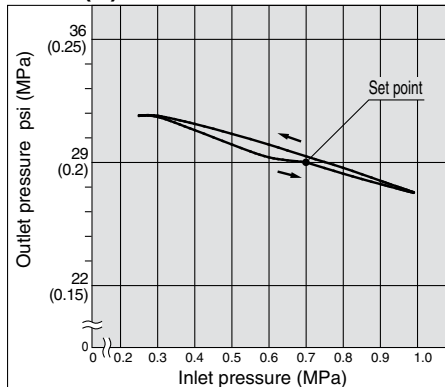
**AW20(K)-B**



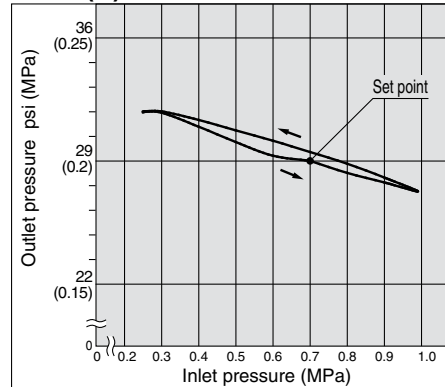
**AW30(K)-B**



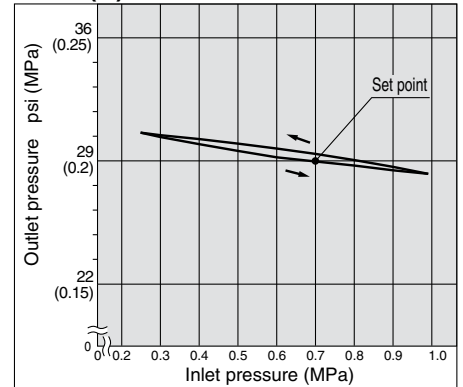
**AW40(K)-B**



**AW40(K)-06-B**

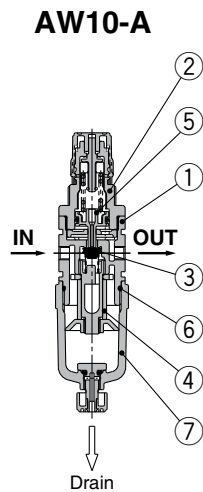


**AW60(K)-B**



psi (MPa) -- 29 (0.2), 44 (0.3), 58 (0.4), 73 (0.5), 87 (0.6), 102 (0.7), 116 (0.8), 131 (0.9), 145 (1.0)

**Construction**



**Component Parts**

| No. | Description | Material      | Color |
|-----|-------------|---------------|-------|
| 1   | Body        | Zinc die-cast | White |
| 2   | Bonnet      | Polyacetal    | White |

**Replacement Parts**

| No. | Description     | Material         | Part no.    |
|-----|-----------------|------------------|-------------|
| 3   | Valve           | HNBR             | AR10P-090S  |
| 4   | Filter element  | Non-woven fabric | AF10P-060S  |
| 5   | Piston assembly | Polyacetal       | AR10P-150AS |
| 6   | Bowl O-ring     | NBR              | C1SFP-260S  |
| 7   | Bowl assembly   | Polycarbonate    | C1SF-A      |

**Working Principle (Filter Regulator with Backflow Function)**

**AW10-A**

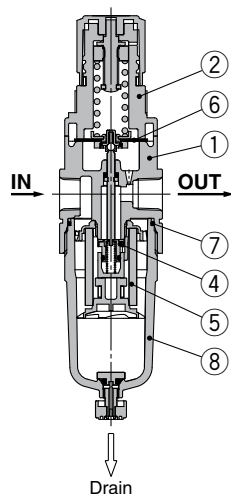


When the inlet pressure is higher than the regulating pressure, the check valve operates as a normal regulator (Figure 1). When the inlet pressure is shut off and exhausted, any inlet pressure applied to the valve ① will be lost. The force for seating the valve ① is the valve spring force ② only. When the valve ① is opened using the outlet force, the outlet pressure will be exhausted at the inlet side (Figure 2). When the set pressure is 0.15 MPa or less, the valve ① may not open due to the valve spring ② force.

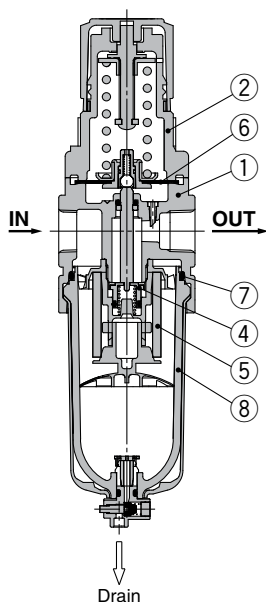
# Series **AW20-B to AW60-B** Series **AW20K-B to AW60K-B**

## Construction

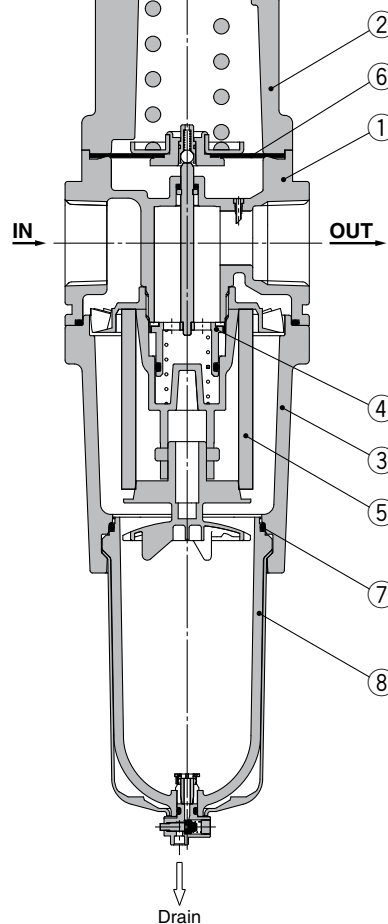
**AW20(K)-B**



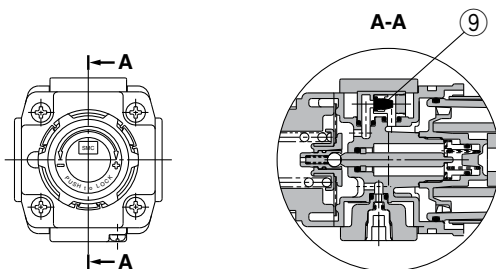
**AW30(K)-B/AW40(K)-B**



**AW60(K)-B**



### **AW20K-B to AW60K-B** (Filter Regulator with Backflow Function)



### Component Parts

| No. | Description    | Material          | Model            | Color |
|-----|----------------|-------------------|------------------|-------|
| 1   | <b>Body</b>    | Aluminum die-cast | AW20-B to AW60-B | White |
| 2   | <b>Bonnet</b>  | Polyacetal        | AW20-B to AW40-B | White |
|     |                | Aluminum die-cast | AW60-B           | White |
| 3   | <b>Housing</b> | Aluminum die-cast | AW60-B           | White |

### Replacement Parts

| No. | Description                             | Material         | Part no.     |                           |                           |              |             |
|-----|-----------------------------------------|------------------|--------------|---------------------------|---------------------------|--------------|-------------|
|     |                                         |                  | AW20(K)-B    | AW30(K)-B                 | AW40(K)-B                 | AW40(K)-06-B | AW60(K)-B   |
| 4   | Valve assembly                          | Brass, HNBR      | AW20P-340AS  | AW30P-340AS               | AW40P-340AS               |              | AW60P-090AS |
| 5   | Filter element                          | Non-woven fabric | AF20P-060S   | AF30P-060S                | AF40P-060S                |              | AW60P-060S  |
| 6   | Diaphragm assembly                      | Weatherable NBR  | AR20P-150AS  | AR30P-150AS               | AR40P-150AS               |              | AR50P-150AS |
| 7   | Bowl seal                               | NBR              | C2SFP-260S   | C32FP-260S                | C42FP-260S                |              |             |
| 8   | Bowl assembly <sup>Note 1)</sup>        | Polycarbonate    | C2SF-A       | C3SF-A <sup>Note 2)</sup> | C4SF-A <sup>Note 2)</sup> |              |             |
| 9   | Check valve assembly <sup>Note 3)</sup> | —                | AR23KP-020AS |                           |                           |              |             |

Note 1) Bowl assembly includes the bowl O-ring.

Please consult with SMC separately for psi and °F unit display specifications.

Note 2) Bowl assembly for the AW30(K)-B to AW60(K)-B models comes with a bowl guard (Material: Polycarbonate).

Note 3) Check valve assembly is applicable for a filter regulator with backflow function (AW20(K)-B to AW60(K)-B) only.

Assembly of a check valve cover, check valve body assembly and 2 mounting screws

## Working Principle (Filter Regulator with Backflow Function)

### AW20K-B to AW60K-B

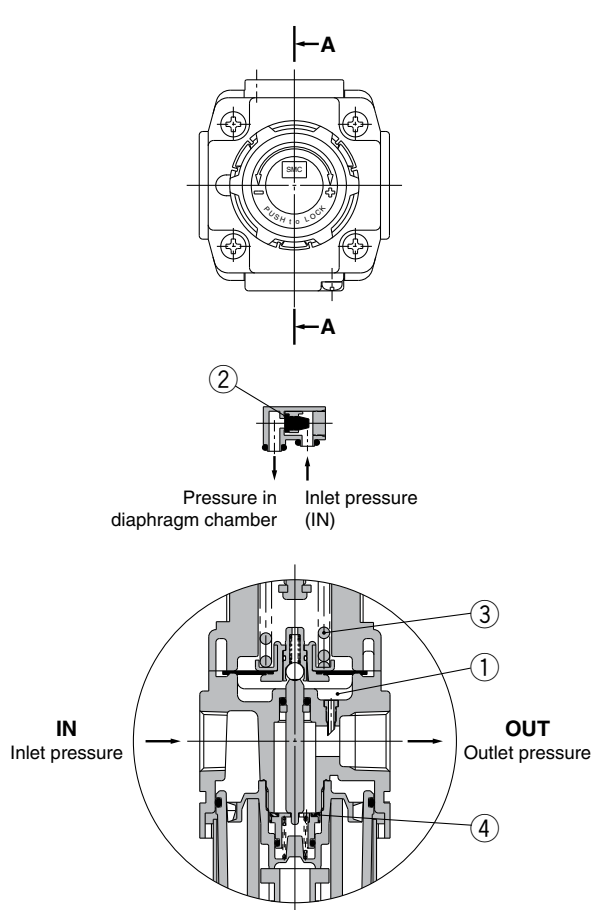


Figure 1 Normal

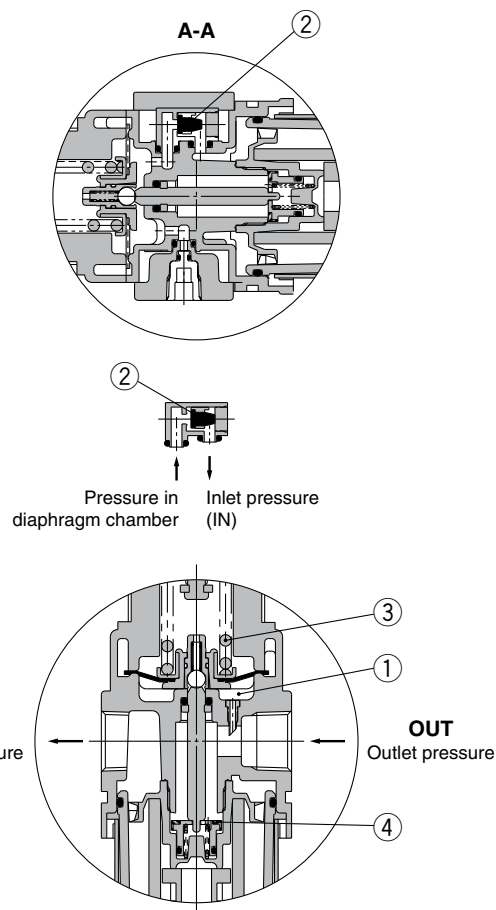


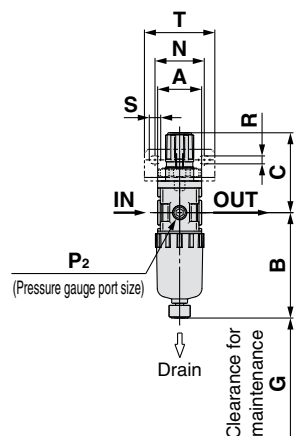
Figure 2 Backflow

When the inlet pressure is higher than the regulating pressure, the check valve ② closes and operates as a normal regulator (Figure 1). When the inlet pressure is shut off and released, the check valve ② opens and the pressure in the diaphragm chamber ① is released into the inlet side (Figure 2). This lowers the pressure in the diaphragm chamber ① and the force generated by the pressure regulator spring ③ lifts the diaphragm. The valve ④ opens through the stem, and the outlet pressure is released to the inlet side (Figure 2).

**Series AW210-A**  
**Series AW20-B to AW60-B**  
**Series AW20K-B to AW60K-B**

**Dimensions**

**AW10-A**



Panel fitting dimensions

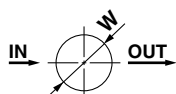
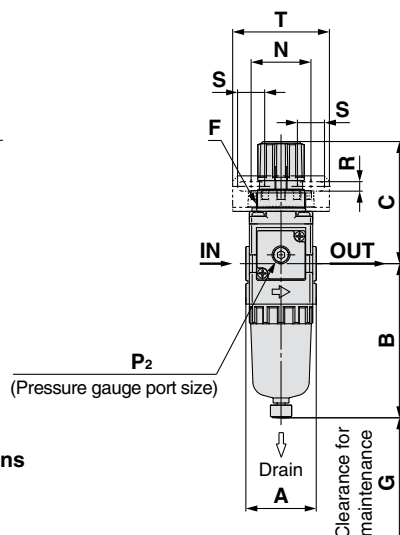


Plate thickness  
 AW10-A: Max. 3.5

**AW20(K)-B**



Panel fitting dimensions

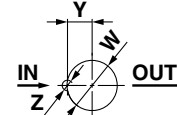
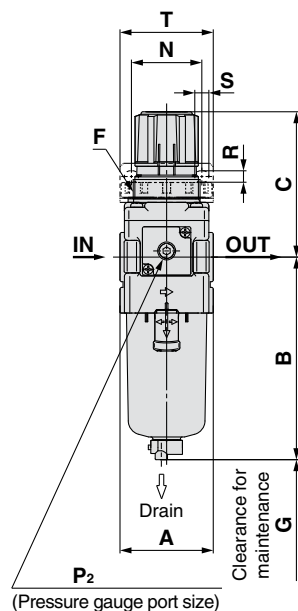


Plate thickness  
 AW20(K)-B: Max. 3.5

**AW30(K)-B to AW40(K)-06-B**



Panel fitting dimensions

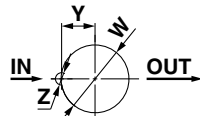
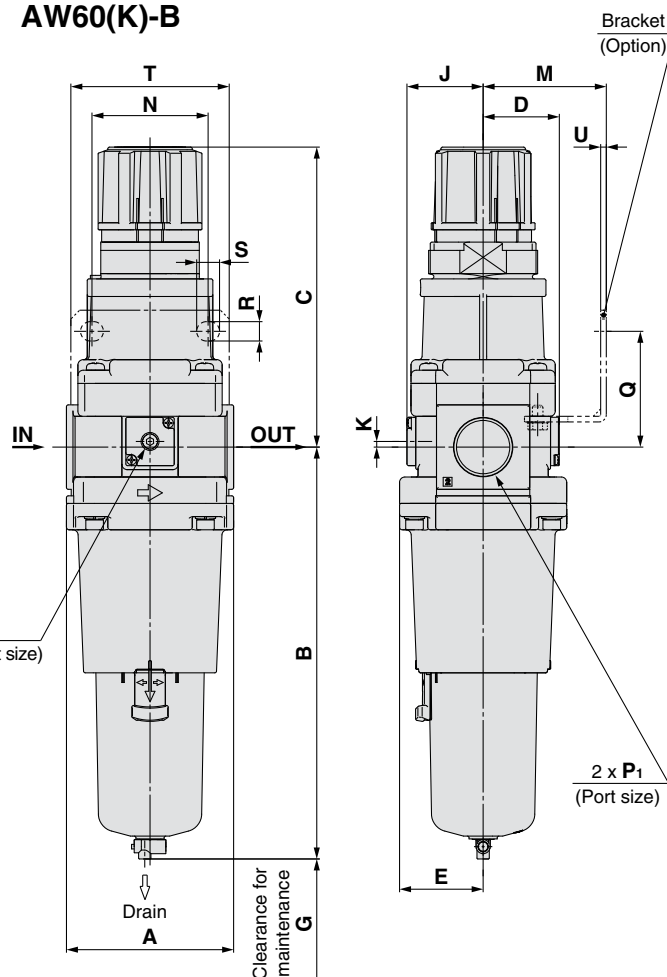
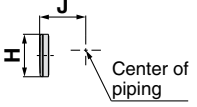
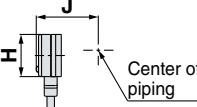
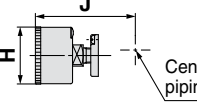


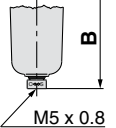
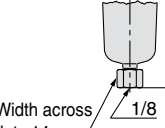
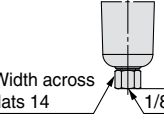
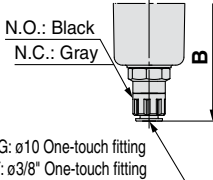
Plate thickness  
 AW30(K)-B: Max. 3.5  
 AW40(K)-B: Max. 5

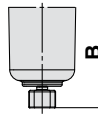
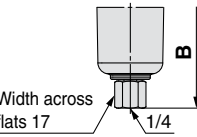
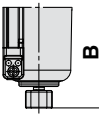
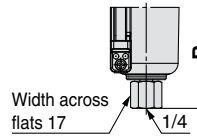
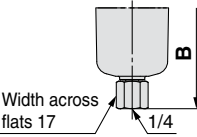
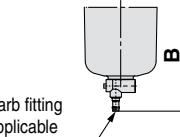
**AW60(K)-B**



**Filter Regulator Series AW210-A**  
**Filter Regulator Series AW20-B to AW60-B**  
**Filter Regulator with Backflow Function Series AW20K-B to AW60K-B**

| Option     | Square embedded type pressure gauge                                               | Digital pressure switch                                                           | Round type pressure gauge                                                          | Round type pressure gauge (with color zone) |
|------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|
| Dimensions |  |  |  |                                             |

| Applicable model                      | AW30(K)-B to AW60(K)-B                                                            |                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Optional/Semi-standard specifications | With auto drain (N.C.)                                                            | Metal bowl                                                                        | With drain guide                                                                  | Metal bowl with drain guide                                                        | With auto drain (N.O./N.C.)                                                         |
| Dimensions                            |  |  |  |  |  |

| Applicable model                      | AW30(K)-B to AW60(K)-B                                                             |                                                                                    |                                                                                    |                                                                                     |                                                                                      |                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Optional/Semi-standard specifications | Metal bowl                                                                         | Metal bowl with drain guide                                                        | Metal bowl with level gauge                                                        | Metal bowl with level gauge, with drain guide                                       | With drain guide                                                                     | Drain cock with barb fitting                                                         |
| Dimensions                            |  |  |  |  |  |  |

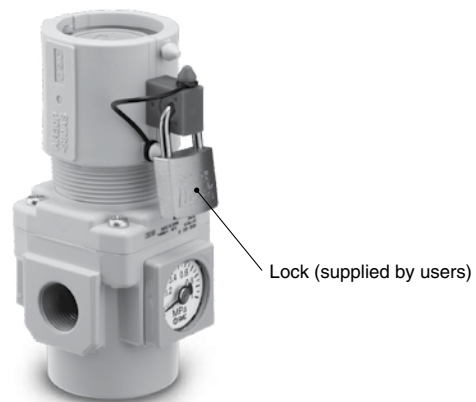
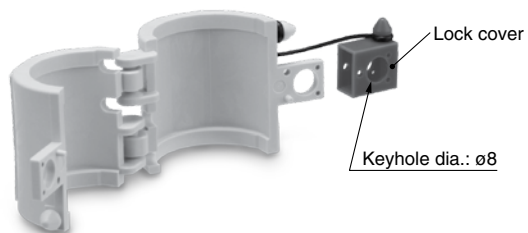
| Model        | Standard specifications |                |    |       |                     |      |      |           |    |      |     | Optional specifications    |      |                         |      |                           |      |                                             |      |
|--------------|-------------------------|----------------|----|-------|---------------------|------|------|-----------|----|------|-----|----------------------------|------|-------------------------|------|---------------------------|------|---------------------------------------------|------|
|              |                         |                |    |       |                     |      |      |           |    |      |     | Square type pressure gauge |      | Digital pressure switch |      | Round type pressure gauge |      | Round type pressure gauge (with color zone) |      |
|              | P <sub>1</sub>          | P <sub>2</sub> | A  | B     | C <sup>(Note)</sup> | D    | E    | F         | G  | J    | K   | H                          | J    | H                       | J    | H                         | J    | H                                           | J    |
| AW10-A       | M5 x 0.8                | 1/16           | 25 | 59.9  | 47.4                | 12.5 | —    | M18 x 1   | 25 | 12.5 | —   | —                          | —    | —                       | —    | —                         | —    | —                                           | —    |
| AW20(K)-B    | 1/8, 1/4                | 1/8            | 40 | 87.6  | 72.4                | 26   | —    | M28 x 1   | 40 | 26   | 5   | □28                        | 27   | □27.8                   | 37.5 | ø37.5                     | 62.5 | ø37.5                                       | 63.5 |
| AW30(K)-B    | 1/4, 3/8                | 1/8            | 53 | 115.1 | 85.6                | 29.4 | 30   | M38 x 1.5 | 55 | 29.4 | 3.5 | □28                        | 30   | □27.8                   | 40.9 | ø37.5                     | 66.9 | ø37.5                                       | 67.9 |
| AW40(K)-B    | 1/4, 3/8, 1/2           | 1/8            | 70 | 147.1 | 91.7                | 37.3 | 38.4 | M42 x 1.5 | 80 | 37.3 | 1.5 | □28                        | 38.4 | □27.8                   | 48.8 | ø42.5                     | 75.7 | ø42.5                                       | 75.7 |
| AW40(K)-06-B | 3/4                     | 1/8            | 75 | 149.1 | 93.2                | 37.3 | 38.4 | M42 x 1.5 | 80 | 37.3 | 1.2 | □28                        | 38.4 | □27.8                   | 48.8 | ø42.5                     | 75.7 | ø42.5                                       | 75.7 |
| AW60(K)-B    | 1                       | 1/8            | 95 | 234.1 | 175.5               | 47.5 | —    | —         | 20 | 47.5 | 3.2 | □28                        | 44.3 | □27.8                   | 61.3 | ø42.5                     | 80.8 | ø42.5                                       | 80.8 |

| Model        | Optional specifications |    |      |     |      |    |     |             |      |    |   |       | Semi-standard specifications |                      |                     |               |                                |                                |                                                  |  |
|--------------|-------------------------|----|------|-----|------|----|-----|-------------|------|----|---|-------|------------------------------|----------------------|---------------------|---------------|--------------------------------|--------------------------------|--------------------------------------------------|--|
|              | Bracket mount           |    |      |     |      |    |     | Panel mount |      |    |   |       | With<br>auto drain           | With<br>barb fitting | With<br>drain guide | Metal<br>bowl | Metal bowl<br>with drain guide | Metal bowl<br>with level gauge | Metal bowl with level gauge,<br>with drain guide |  |
|              | M                       | N  | Q    | R   | S    | T  | U   | V           | W    | Y  | Z | B     | B                            | B                    | B                   | B             | B                              | B                              |                                                  |  |
| AW10-A       | 25                      | 28 | 30   | 4.5 | 6.5  | 40 | 2   | 18          | 18.5 | —  | — | 77.9  | —                            | —                    | 59.3                | —             | —                              | —                              |                                                  |  |
| AW20(K)-B    | 30                      | 34 | 43.9 | 5.4 | 15.4 | 55 | 2.3 | 24.7        | 28.5 | 14 | 6 | 104.9 | —                            | 91.4                 | 87.4                | 93.9          | —                              | —                              |                                                  |  |
| AW30(K)-B    | 41                      | 40 | 45.8 | 6.5 | 8    | 53 | 2.3 | 31.1        | 38.5 | 19 | 7 | 156.8 | 123.6                        | 121.9                | 117.6               | 122.1         | 137.6                          | 142.1                          |                                                  |  |
| AW40(K)-B    | 50                      | 54 | 54   | 8.5 | 10.5 | 70 | 2.3 | 35.5        | 42.5 | 21 | 7 | 186.9 | 155.6                        | 153.9                | 149.6               | 154.1         | 169.6                          | 174.1                          |                                                  |  |
| AW40(K)-06-B | 50                      | 54 | 55.5 | 8.5 | 10.5 | 70 | 2.3 | 37          | 42.5 | 21 | 7 | 188.9 | 157.6                        | 155.9                | 151.6               | 156.1         | 171.6                          | 176.1                          |                                                  |  |
| AW60(K)-B    | 70                      | 66 | 65.8 | 11  | 13   | 90 | 3.2 | —           | —    | —  | — | 273.9 | 240.9                        | 242.6                | 236.6               | 241.1         | 256.6                          | 261.1                          |                                                  |  |

Note) The dimension of C is the length when the filter regulator knob is unlocked.

# Option Knob Cover

Prevents careless knob operation.



| Part no.           | Applicable model                                |
|--------------------|-------------------------------------------------|
| <b>AR20P-580AS</b> | AC20□-B, AR20(K)-B, AW20(K)-B                   |
| <b>AR25P-580AS</b> | AC25□-B, AR25(K)-B                              |
| <b>AR30P-580AS</b> | AC30□-B, AR30(K)-B, AW30(K)-B                   |
| <b>AR40P-580AS</b> | AC40□(-06)-B, AR40(K)(-06)-B,<br>AW40(K)(-06)-B |